

1] Determine which of the following expression are allowed in index notation, if they are not allowed state why (a , b , c , d , and e are arbitrary quantities) (10 points)

- (a) $a = b_i c_{ij} d_j$
- (b) $a = b_i c_i + d_j$
- (c) $a_i = \delta_{ij} b_i + c_i$
- (d) $a_k = b_i c_{ki}$
- (e) $a_k = b_k c + d_i e_{ik}$
- (f) $a_i = b_i + c_{ij} d_{ji} e_i$
- (g) $a_l = \epsilon_{ijk} b_j c_k$
- (h) $a_{ij} = b_{ji}$
- (i) $a_{ij} = b_i c_j + e_{jk}$
- (j) $a_{kl} = b_i c_{ki} d_l + e_{ki}$

2] For three spatial dimensions, rewrite the following expressions in index notation and evaluate or simplify them using the values or parameters given, and the definitions of δ_{ij} and ϵ_{ijk} wherever possible. In parts b) through e), $\vec{x}$ is the position vector, with components x_i .

- a) $\vec{b} \cdot \vec{c}$, where $\vec{b} = (1, 4, 17)$ and $\vec{c} = (-4, -3, 1)$.
- b) $(\vec{u} \cdot \vec{\nabla}) \vec{x}$, where $\vec{u}$ is a vector with components u_i
- c) $\vec{\nabla} \phi$, where $\phi = \vec{h} \cdot \vec{x}$ and $\vec{h}$ is a constant vector with components h_i
- d) $\vec{\nabla} \times \vec{u}$, where $\vec{u} = \vec{\Omega} \times \vec{x}$ and $\vec{\Omega}$ is a constant vector with components Ω_i
- e) $\vec{C} \cdot \vec{x}$, where

$$\vec{C} = \begin{Bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{Bmatrix}$$

3] Prove that the following expression is true using Cartesian index notation without the use of vector identities (if you want to use a vector identity you must first show it is true using index notation).

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$$

4] Prove that the following expressions are true using Cartesian index notation without the use of vector identities (if you want to use a vector identity you must first show it is true using index notation).

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{c} \cdot (\vec{d} \times \vec{a})] \vec{b} - [\vec{c} \cdot (\vec{d} \times \vec{b})] \vec{a}$$

And

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \cdot (\vec{b} \times \vec{d})] \vec{c} - [\vec{a} \cdot (\vec{b} \times \vec{c})] \vec{d}$$